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Process-induced forming of oxide RRAM

Abstract

Embodiments of present invention provide a method of forming a resistive random-access memory (RRAM). The method includes forming a dielectric layer on top of a supporting structure, wherein the dielectric layer has a bottom electrode embedded therein; forming an oxide layer on top of the bottom electrode; treating the oxide layer in a plasma environment; forming a top electrode on top of the oxide layer; forming a first interlevel-dielectric (ILD) layer on top of the top electrode; forming a via contact and a first metal layer in the first ILD layer, wherein the first metal layer is in contact with the top electrode through the via contact; forming a capping layer on top of the first metal layer through a plasma-enhanced deposition process; and causing formation of one or more filaments in the oxide layer during the plasma-enhanced deposition process. A structure of the RRAM is also provided.

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Background/Summary

BACKGROUND
(1) The present application relates to manufacturing of semiconductor integrated circuits. More particularly, it relates to an oxide resistive random-access memory device and the electroforming of the device.
(2) Resistive random-access memory (RRAM) or RRAM device is widely considered as a promising technology for use as electronic synapse devices or memristors in neuromorphic computing as well as high-density and high-speed non-volatile memory applications. In

neuromorphic computing applications, a RRAM device may be used as a connection (synapse) between a pre-neuron and a post-neuron, representing the connection weight in the form of device conductance. Multiple pre-neurons and post-neurons may be connected through a crossbar of RRAM device array, which naturally expresses a fully connected neural network.
(3) For oxide RRAM device, a current conducting filament is formed within the switching layer of the device during an initial operation known as electroforming. Electroforming of oxide RRAM device is similar to dielectric break-down and it typically follows Poisson area scaling. In other words, the smaller the RRAM device size is, the higher the needed forming voltage becomes in the electroforming process. For example, the forming voltage of an oxide RRAM device at sub-um dimension is typically larger than 2V and this forming voltage increases as the dimension of the device further scales down. On the other hand, in state-of-the-art CMOS technologies, a single metal-oxide-semiconductor field-effect-transistor (MOSFET) may not be able to support such a large forming voltage for the purpose of electroforming the device. One compromise may thus be to supply such forming voltage by forming multiple MOSFETs that are stacked together, which nevertheless introduces significant penalty on device footprint.

SUMMARY

(4) Embodiments of present invention provide a semiconductor structure that includes a resistive random-access memory (RRAM). The RRAM includes a bottom electrode, an oxide layer on top of the bottom electrode, and a top electrode on top of the oxide layer, wherein the oxide layer includes hydrogen; a first metal layer in a first interlevel-dielectric (ILD) layer and above the RRAM, the first metal layer being in contact with the top electrode of the RRAM through a via contact; and a capping layer directly on top of the first metal layer, wherein the capping layer is a layer of carbon-hydrogen containing silicon-nitride.
(5) In one embodiment, the oxide layer is a layer of hafnium-oxide (HfO), tantalum-oxide (TaO), titanium-oxide (TiO) or a combination thereof.
(6) In another embodiment, the oxide layer contains hydrogen, and the hydrogen in the oxide layer has a concentration level that is less than 10.sup.20 atoms/cm.sup.3.
(7) In one embodiment, the top electrode has a first size and the bottom electrode has a second size, and the first size is larger than the second size.
(8) In another embodiment, the bottom electrode is embedded in a dielectric layer, the bottom electrode and the dielectric layer being directly on top of a second metal layer and the second metal layer being embedded in a second ILD layer.
(9) In one embodiment, the top electrode and the oxide layer of the RRAM are encapsulated by an encapsulation layer, and the via contact is in direct contact with the top electrode through at least the encapsulation layer.
(10) Embodiments of present invention also provide a method of forming a semiconductor structure that contains at least one resistive random-access memory. The method includes forming a dielectric layer on top of a supporting structure, wherein the dielectric layer has a bottom electrode embedded therein; forming an oxide layer on top of the bottom electrode; forming a top electrode on top of the oxide layer; forming a first interlevel-dielectric (ILD) layer on top of the top electrode; forming a via contact and a first metal layer in the first ILD layer, wherein the first metal layer is in contact with the top electrode through the via contact; forming a capping layer on top of the first metal layer through a plasma-enhanced chemical-vapor-deposition (PECVD) process; and causing formation of one or more filaments in the oxide layer during the PECVD process of forming the capping layer.
(11) In one embodiment, the oxide layer is a layer of hafnium-oxide (HfO), tantalum-oxide (TaO), titanium-oxide (TiO), or a combination thereof.
(12) According to one embodiment, the method further includes, before forming the top electrode, subjecting the oxide layer to a plasma environment that contains hydrogen, thereby causing the oxide layer to contain hydrogen with a concentration level that is less than 10.sup.20

atoms/cm.sup.3 and to have oxygen vacancies.
(13) In one embodiment, causing the formation of one or more filaments includes funneling free electrons to the oxide layer during the plasma-enhanced deposition process and using the oxygen vacancies as precursors to form the one or more filaments.
(14) In another embodiment, the capping layer is a carbon-hydrogen containing silicon-nitride layer formed through the PECVD process.
(15) In one embodiment, the capping layer is formed in the PECVD process in an environment that contains precursors of at least silicon, nitrogen, carbon, and hydrogen.
(16) According to one embodiment, the method further includes forming an encapsulation layer to cover the oxide layer and the top electrode.
(17) In one embodiment, the encapsulation layer is a conformal dielectric layer.
(18) In another embodiment, the via contact is made through the encapsulation layer to be in direct contact with the top electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention will be understood and appreciated more fully from the following detailed description of embodiments of present invention, taken in conjunction with accompanying drawings of which:
(2) FIGS. **1-11** are demonstrative illustrations of cross-sectional views of a resistive random-access memory device in a series of steps of manufacturing thereof according to embodiments of present invention; and
(3) FIG. **12** is a demonstrative illustration of a flow-chart of a method of manufacturing a resistive random-access memory device according to embodiments of present invention.
(4) It will be appreciated that for simplicity and clarity purpose, elements shown in the drawings have not necessarily been drawn to scale. Further, and if applicable, in various functional block diagrams, two connected devices and/or elements may not necessarily be illustrated as being connected. In some other instances, grouping of certain elements in a functional block diagram may be solely for the purpose of description and may not necessarily imply that they are in a single physical entity, or they are embodied in a single physical entity.
DETAILED DESCRIPTION
(5) In the below detailed description and the accompanying drawings, it is to be understood that various layers, structures, and regions shown in the drawings are both demonstrative and schematic illustrations thereof that are not drawn to scale. In addition, for the ease of explanation, one or more layers, structures, and regions of a type commonly used to form semiconductor devices or structures may not be explicitly shown in a given illustration or drawing. This does not imply that any layers, structures, and regions not explicitly shown are omitted from the actual semiconductor structures. Furthermore, it is to be understood that the embodiments discussed herein are not limited to the particular materials, features, and processing steps shown and described herein. In particular, with respect to semiconductor processing steps, it is to be emphasized that the descriptions provided herein are not intended to encompass all of the processing steps that may be required to form a functional semiconductor integrated circuit device. Rather, certain processing steps that are commonly used in forming semiconductor devices, such as, for example, wet cleaning and annealing steps, are purposefully not described herein for economy of description.
(6) It is to be understood that the terms “about” or “substantially” as used herein with regard to thicknesses, widths, percentages, ranges, etc., are meant to denote being close or approximate to, but not exactly. For example, the term “about” or “substantially” as used herein implies that a small margin of error may be present such as, by way of example only, 1% or less than the stated amount.

Likewise, the terms “on”, “over”, or “on top of” that are used herein to describe a positional relationship between two layers or structures are intended to be broadly construed and should not be interpreted as precluding the presence of one or more intervening layers or structures.
(7) To provide spatial context to different structural orientations of the semiconductor structures shown in the drawings, XYZ Cartesian coordinates may be provided in some of the drawings. The terms “vertical” or “vertical direction” or “vertical height” as used herein denote a Z-direction of the Cartesian coordinates shown in the drawings, and the terms “horizontal” or “horizontal direction” or “lateral direction” as used herein denote an X-direction and/or a Y-direction of the Cartesian coordinates shown in the drawings.
(8) Moreover, although various reference numerals may be used across different drawings, the same or similar reference numbers are used throughout the drawings to denote the same or similar features, elements, or structures, and thus detailed explanations of the same or similar features, elements, or structures may not be repeated for each of the drawings for economy of description. Labelling for the same or similar elements in some drawings may be omitted as well in order not to overcrowd the drawings.
(9) FIGS. **1-11** are demonstrative illustrations of cross-sectional views of a resistive random-access memory (RRAM) device **10** in a series of steps of manufacturing thereof according to embodiments of present invention. More particularly, as is illustrated in FIG. **1**, embodiments of present invention provide receiving a supporting structure such as, for example, a back-end-of-line (BEOL) structure that includes a dielectric layer **120** with a metal layer **121** embedded therein. The metal layer **121** may be, for example, a M2 or M3 layer and may be a layer of copper (Cu), cobalt (Co), tungsten (W), aluminum (Al), ruthenium (Ru), or other conductive materials. The dielectric layer **120** may be on top of another dielectric layer **110**.
(10) Embodiments of present invention further provide forming one or more bottom electrodes on top of the metal layer **121**. Here, for the purpose of description, a first bottom electrode **211** and a second bottom electrode **212** are demonstratively illustrated in FIG. **1**. The first bottom electrode **211** and the second bottom electrode **212** are formed directly on top of the metal layer **121** for a first RRAM **241** and a second RRAM **242** respectively. However, embodiments of present invention are not limited in this aspect. For example, fewer or greater bottom electrodes may be formed in forming a corresponding fewer or greater number of RRAMs. The first and second bottom electrodes **211** and **212** may include, for example, titanium-nitride (TiN) or tantalum-nitride (TaN) and may be formed to be embedded in a dielectric layer **210**. In one embodiment, the dielectric layer **210** may include, for example, silicon-nitride (SiN) or silicon-carbonitride (SiNC).
(11) More specifically, in forming the first and second bottom electrodes **211** and **212**, one embodiment of present invention may provide forming the dielectric layer **210** on top of the metal layer **121** and the dielectric layer **120**. One or more openings may then be created in the dielectric layer **210**, through a lithographic patterning and etching process, to expose the metal layer **121**. Next, conductive materials such as TiN or TaN may be deposited through, for example, a chemical-vapor-deposition (CVD) process, a physical-vapor-deposition (PVD) process, or an atomic-layer-deposition (ALD) process in the openings to form the bottom electrodes **211** and **212**. Following the deposition of the conductive material, a chemical-mechanic-polishing (CMP) process may be applied, for example, to planarize top surfaces of the bottom electrodes **211** and **212**.
(12) As is illustrated in FIG. **2**, embodiments of present invention further provide forming a blanket oxide layer **220** on top of the bottom electrodes **211** and **212** and the dielectric layer **210**. The blanket oxide layer **220** may be a layer of, for example, hafnium-oxide (HfOx), tantalum-oxide (TaOx), titanium-oxide (TiOx), or a combination thereof and may be formed to have a thickness ranging from about 3 nm to about 20 nm. Here, x represents an integer such as 1, 2, etc. For simplicity, sometimes hafnium-oxide, tantalum-oxide, and titanium-oxide may simply be represented by HfO, TaO, and TiO respectively. The formation of the blanket oxide layer **220** may be made through, for example, a deposition process such as, for example, a CVD, PVD, or ALD

process, although other currently existing or future developed means or processes may be used as well.
(13) Embodiments of present invention may further include subjecting the blanket oxide layer **220** to a hydrogen (H.sub.2) based plasma treatment. This plasma treatment creates oxygen vacancies within the blanket oxide layer **220** that may serve as precursors for oxide breakdown during a downstream processing of forming filament according to embodiment of present invention. The hydrogen-based plasma treatment may also leave residual hydrogen atoms in the blanket oxide layer **220** after treatment. For example, the concentration level of hydrogen atoms in the blanket oxide layer **220**, also known as hydrogen impurity, may be less than, for example, 10.sup.20 atoms/cm.sup.3 and in one embodiment may be between 10.sup.18 atoms/cm.sup.3 and 10.sup.20 atoms/cm.sup.3 after the plasma treatment.
(14) As is illustrated in FIG. **3**, embodiments of present invention provide forming a top electrode layer **230** on top of the blanket oxide layer **220** and a hard mask layer **310** on top of the top electrode layer **230**. More specifically, the top electrode layer **230** may be formed through a deposition process, such as a CVD, PVD or ALD process, to have a thickness ranging from about 10 nm to about 30 nm. The top electrode layer **230** may be made of or include TiN or TaN and in one embodiment may be made of a same material as the bottom electrodes **211** and **212**. Following the formation of the top electrode layer **230**, the hard mask layer **310** may be formed on top thereof. The hard mask layer **310** may be a layer of dielectric material such as SiN and may be formed to have a thickness ranging from about 20 nm to about 60 nm.
(15) As is illustrated in FIG. **4**, embodiments of present invention provide patterning the hard mask layer **310** into a hard mask **311** through, for example, a lithographic patterning process and a subsequent directional etching process such as a reactive-ion-etching (RIE) process. The formed hard mask **311** represents a pattern of a top electrode and an oxide layer underneath thereof. In the embodiment demonstratively illustrated in FIG. **4**, the hard mask **311** represents a pattern of the top electrode and the oxide layer to be shared by the first RRAM **241** and the second RRAM **242**. However, embodiments of present invention are not limited in this aspect. In one embodiment, the hard mask **311** may represent a pattern of a top electrode of only one RRAM. For example, the hard mask **311** may be formed to pattern a top electrode of the first RRAM **241** or alternatively of the second RRAM **242**.
(16) Embodiments of present invention further provide etching the top electrode layer **230** and the blanket oxide layer **220** into a top electrode **231** and an oxide layer **221**, using the hard mask **311** in an anisotropic and/or directional etching process such as a RIE process. In the illustrated embodiment, the top electrode **231** and the oxide layer **221** underneath thereof cover both the first bottom electrode **211** of the first RRAM **241** and the second bottom electrode **212** of the second RRAM **242**. In one embodiment, in a top view, the top electrode **231** and the oxide layer **221** may have a first size and the first bottom electrode **211** and/or the second bottom electrode **212** may have a second size. The first size may be larger than the second size.
(17) As is illustrated in FIG. **5**, embodiments of present invention provide forming an encapsulation layer **320** covering the RRAM device **10**. More specifically, the encapsulation layer **320** may cover a top surface of the hard mask **311**, sidewall surfaces of the hard mask **311**, the top electrode **231**, the oxide layer **221**, and a portion of the top surface of the dielectric layer **210** that is not covered by the oxide layer **221**. In one embodiment, the encapsulation layer **320** may be a conformal dielectric layer or a conformal dielectric liner and may be deposited or formed to have a thickness ranging from about 5 nm to about 20 nm. The encapsulation layer **320** may include SiN, SiNC, or other suitable dielectric materials.
(18) As is illustrated in FIG. **6**, embodiments of present invention provide forming a first interlevel-dielectric (ILD) layer **410** covering the RRAM device **10**. More specifically, the first ILD layer **410** may be a silicon-dioxide (SiO.sub.2) layer, a porous SiO.sub.2 layer, or other low-k dielectric material layer and may be formed through a deposition process, such as a CVD, PVD, or ALD

process, directly on top of the encapsulation layer **320**. The first ILD layer **410** may be formed to have a thickness ranging from about 300 nm to about 500 nm. A CMP process may be used to planarize the top surface of the first ILD layer **410**.

(19) As is illustrated in FIG. **7**, embodiments of present invention provide forming a via hole **421** in the first ILD layer **410** through a lithographic patterning and etching process. For example, an etch mask may first be formed to cover most of the first ILD layer **410** except a surface area where the via hole **421** is to be made. Subsequently, an anisotropic or directional etching process may be used to etch the portion of the first ILD layer **410** that is exposed by the etch mask resulting the via hole **421**. This etching process may be a selective etching process and may stop at the encapsulation layer **320**. The size of the via hole **421** may typically represent the size of a via contact, to be formed later, to contact the top electrode **231**.

(20) As is illustrated in FIG. **8**, embodiments of present invention provide forming a trench opening **422** above the via hole **421** and to overlap with the via hole **421**. The trench opening **422** may be formed for forming a metal layer, such as a M3 or M4 layer, in the first ILD layer **410**, and thus may be formed to have a depth that is less than that of the via hole **421** and definitely less than a thickness of first ILD layer **410**. The trench opening **422** may be formed or created through a lithographic patterning and etching process, similar to that of forming the via hole **421**.

(21) As is illustrated in FIG. **9**, embodiments of present invention provide performing another selective etching process that selectively etches the encapsulation layer **320** and the hard mask **311** underneath thereof. In other words, this selective etching process may not substantially etch the first ILD layer **410**. Instead, in this selective etching process, the first ILD layer **410** may be used as an etch mask and the pattern of the via hole **421** may be transferred to the encapsulation layer **320** and the hard mask **311**. In other words, the etching process may create a modified via hole **423** that goes through the encapsulation layer **320** and the hard mask **311** and selectively stops at the top electrode **231**, thereby exposing a top surface of the top electrode **231**.

(22) As is illustrated in FIG. **10**, embodiments of present invention provide filling the modified via hole **423** with a conductive material, for example through a CVD, PVD, or ALD deposition process, to form a via contact **431** and filling the trench opening **422** above the modified via hole **423**, with a same or different conductive material, to form a metal layer **432**. In one embodiment, the conductive material may be copper (Cu), cobalt (Co), tungsten (W), aluminum (Al), ruthenium (Ru), or other conductive materials. After the deposition of the conductive material, a CMP process may be used to planarize a top surface of the metal layer **432**. The metal layer **432** may be embedded in the first ILD layer **410** and may be in contact with the top electrode **231** through the via contact **431**. In other words, the via contact **431** is in direct contact with the top electrode **231** through at least the encapsulation layer **320** and thus the metal layer **432** is conductively connected to the top electrode **231**.

(23) As is illustrated in FIG. **11**, embodiments of present invention provide forming a capping layer **510** directly on top of the metal layer **432**. The capping layer **510** may also be formed on top of the first ILD layer **410**. In one embodiment, the capping layer **510** may be formed through a plasma-enhanced chemical-vapor-deposition (PECVD) process of depositing a silicon-nitride layer in an environment that contains carbon-hydrogen. As a result, the capping layer **510** formed thereby may be a carbon-hydrogen containing silicon-nitride. In one embodiment, the capping layer **510** may be a layer of NBLoK film, which may be a material represented by SiC.sub.xN.sub.yH.sub.z where x, y, z are integers such as 1, 2, etc. For simplicity, the NBLok film may be represented sometimes simply as SiCNH.

(24) During the PECVD process, free electrons in the plasma ambient environment may bombard the surface of the metal layer **432**. Since the total surface area of the metal layer **432** is generally significantly larger than that of the first RRAM **241** and/or the second RRAM **242**, the metal layer **432** serves as an antenna that collects the free electrons from the plasma ambient and subsequently funnels them to the first RRAM **241** and/or the second RRAM **242** of the RRAM device **10**. At one

point, the density of the free electrons may become sufficiently high to cause electro-forming when they reach the area of the RRAM device.

(25) This electro-forming process generally only happens during the initial stage of the deposition step of the capping layer **510** by virtue of the self-limiting mechanism or effect of the capping layer **510** formed on top of the device. More particularly, the deposited capping layer **510** blocks excessive plasma bombardment, thereby avoiding formation of excessive filament that facilitates controlled filament formation.

(26) FIG. **12** is a demonstrative illustration of a flow-chart of a method of manufacturing a RRAM device according to embodiments of present invention. The method includes (**910**) providing a supporting structure, and forming a bottom electrode embedded in a dielectric layer are on top of the supporting structure; (**920**) forming a blanket oxide layer on top of the bottom electrode, and treating the blanket oxide layer with a hydrogen-containing plasma; (**930**) forming a top electrode layer on top of the blanket oxide layer and patterning the top electrode and the blanket oxide layer to form an oxide layer and a top electrode on top thereof; (**940**) covering the oxide layer and the top electrode with an encapsulation layer such as a conformal dielectric layer; (**950**) forming an interlevel-dielectric (ILD) layer on top of the RRAM device and particularly on top of the encapsulation layer, and creating a via hole and a trench opening on top of and overlapping with the via hole, thereby exposing the top electrode; (**960**) filling the via hole and the trench opening to create a metal layer in the ILD layer, the metal layer being in contact with the top electric through a via contact formed in the via hole; and (**970**) forming a carbon-hydrogen containing silicon-nitride capping layer on top of the metal layer in a plasma-enhanced CVD process, wherein the plasma-enhanced CVD process enables the formation of filament in the oxide layer of the RRAM device.

(27) It is to be understood that the exemplary methods discussed herein may be readily incorporated with other semiconductor processing flows, semiconductor devices, and integrated circuits with various analog and digital circuitry or mixed-signal circuitry. In particular, integrated circuit dies can be fabricated with various devices such as field-effect transistors, bipolar transistors, metal-oxide-semiconductor transistors, diodes, capacitors, inductors, etc. An integrated circuit in accordance with the present invention can be employed in applications, hardware, and/or electronic systems. Suitable hardware and systems for implementing the invention may include, but are not limited to, personal computers, communication networks, electronic commerce systems, portable communications devices (e.g., cell phones), solid-state media storage devices, functional circuitry, etc. Systems and hardware incorporating such integrated circuits are considered part of the embodiments described herein. Given the teachings of the invention provided herein, one of ordinary skill in the art will be able to contemplate other implementations and applications of the techniques of the invention.

(28) Accordingly, at least portions of one or more of the semiconductor structures described herein may be implemented in integrated circuits. The resulting integrated circuit chips may be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip may be mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other high-level carrier) or in a multichip package (such as a ceramic carrier that has surface interconnections and/or buried interconnections). In any case the chip may then be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product, such as a motherboard, or an end product. The end product may be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

(29) The descriptions of various embodiments of present invention have been presented for the purposes of illustration and they are not intended to be exhaustive and present invention are not limited to the embodiments disclosed. The terminology used herein was chosen to best explain the principles of the embodiments, practical application or technical improvement over technologies

found in the marketplace, and to enable others of ordinary skill in the art to understand the embodiments disclosed herein. Many modifications, substitutions, changes, and equivalents will now occur to those of ordinary skill in the art. Such changes, modification, and/or alternative embodiments may be made without departing from the spirit of present invention and are hereby all contemplated and considered within the scope of present invention. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the spirit of the invention.

Claims

1. A method of forming a semiconductor structure, the method comprising: forming a dielectric layer on top of a supporting structure, wherein the dielectric layer has a bottom electrode embedded therein; forming an oxide layer on top of the bottom electrode; forming a top electrode on top of the oxide layer; forming an encapsulation layer to cover the oxide layer and the top electrode; forming a first interlevel-dielectric (ILD) layer on top of the top electrode; forming a via contact and a first metal layer in the first ILD layer, wherein the first metal layer is in contact with the top electrode through the via contact; forming a capping layer on top of the first metal layer; and causing formation of one or more filaments in the oxide layer.
2. The method of claim 1, wherein the oxide layer is a layer of hafnium-oxide (HfO), tantalum-oxide (TaO), titanium-oxide (TiO), or a combination thereof; the capping layer is formed through a plasma-enhanced-chemical-vapor-deposition (PECVD) process; and the one or more filaments are formed during the PECVD process.
3. The method of claim 2, further comprising, before forming the top electrode, subjecting the oxide layer to a plasma environment that contains hydrogen, thereby causing the oxide layer to contain hydrogen with a concentration level that is less than 10.sup.20 atoms/cm.sup.3 and to have oxygen vacancies.
4. The method of claim 3, wherein causing the formation of one or more filaments in the oxide layer comprises funneling free electrons to the oxide layer during the PECVD process and using the oxygen vacancies as precursors in forming the one or more filaments.
5. The method of claim 1, wherein the capping layer is a layer of carbon-hydrogen containing silicon-nitride that is formed through a plasma-enhanced-chemical-vapor-deposition process.
6. The method of claim 1, wherein the capping layer is formed in a plasma-enhanced-chemical-vapor-deposition process in an environment that contains precursors of at least silicon, nitrogen, carbon, and hydrogen.
7. The method of claim 1, wherein the encapsulation layer is a conformal dielectric layer.
8. The method of claim 1, wherein the via contact is made through the encapsulation layer to be in direct contact with the top electrode.